Jesse McNamara
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Profile

A hard working and dedicated engineering student seeking an internship to gain industry experience and develop technical and professional skills. With a keen interest in developing proficiencies in electrical hardware design and to contribute to cutting edge technologies and systems. I hope to utilise an internship in these areas to contribute to innovative and meaningful projects.

Education

Bachelor of Engineering (Honours), Mechatronic Engineering *Expected Jun 2024*
Bachelor of Science, Applied Physics

- University of Technology Sydney
- John Heine Memorial Foundation Academic Prize for first year
- Dean's List 2017, 2018, 2019

Work History

UTS Motorsports Electric *July 2018 – Nov 2019*
Vehicle Dynamics Section Lead

- Research and design of suspension kinematics and mechanical systems including push-rod, pull-rod and direct acting
- Lead a small group of engineering students to design and build a FSAE suspension system
- Producing drawings and part/assembly models using SolidWorks and designed for CAM manufacture
- Completed FEA simulation using Ansys and SolidWorks to analyse static loading
- Design and analysis of mechanical parts, specifically wheel assembly, and suspension mechanisms starting from first principles
- Transitioned the team's suspension from direct acting to a pushrod system

University of Technology Sydney *Jun 2019 – Jun 2023*
Tutor/Lab demonstrator

- Presented face-to-face and zoom based classes for introduction to C++ and ROS, marked assignments and provided student feedback
- Developed communication skills and teaching methodologies, gained experience explaining complex ideas clearly with the aid of diagrams/demonstration
- Maintained high level of positive student feedback
- Mentoring students on fundamental skills including mathematics and mechanics

Labyrinth Escape Rooms *Nov 2020 – Present*

Embedded automation

- Designed and assembled 2-layer PCBs using KiCad to interface user input with microcontrollers and serial outputs to facilitate puzzle automation
- Maintain/improve existing automation systems for reliability, including writing error tolerant code and transitioning to more robust electrical connectors
- Added ethernet and CAN communication networks to escape rooms using microcontroller boards to add redundancy to failure and facilitate more complex puzzles

Projects

Fluorescence Imaging Capstone *Jan 2024 – Present*

- Designed 220W LED illumination array using constant current buck drivers for fluorescence excitation. Optimised LED locations for uniform near-field irradiance
- Applied current shunts to facilitate fast LED switching speeds and generate pulses <10us
- Developed aluminium back PCB in KiCad to maximise thermal conduction away from LEDs
- Developed AVR microcontroller PCB with DAC output for LED current control and UART to USB serial converter for computer interface.
- Developed GUI application for gigE machine vision camera using qt5, and interface for USB communications with microcontroller.

Solar System Mechanics Simulation *Mar 2023 – May 2023*

- Created MATLAB model of bodies within the solar system using a 2nd order symplectic numerical integration model.

Skills

Software: KiCad, SolidWorks, LTspice, PSpice, ROS
Languages: C/C++, Python, MATLAB, JS
Hardware: Microcontrollers(AVR, ARM, PIC), Raspberry Pi, UpBoard
Protocols: UART, SPI, I2C, CAN, Ethernet
Equipment: Oscilloscope, Multimeter, Function generator, Logic Analyzer